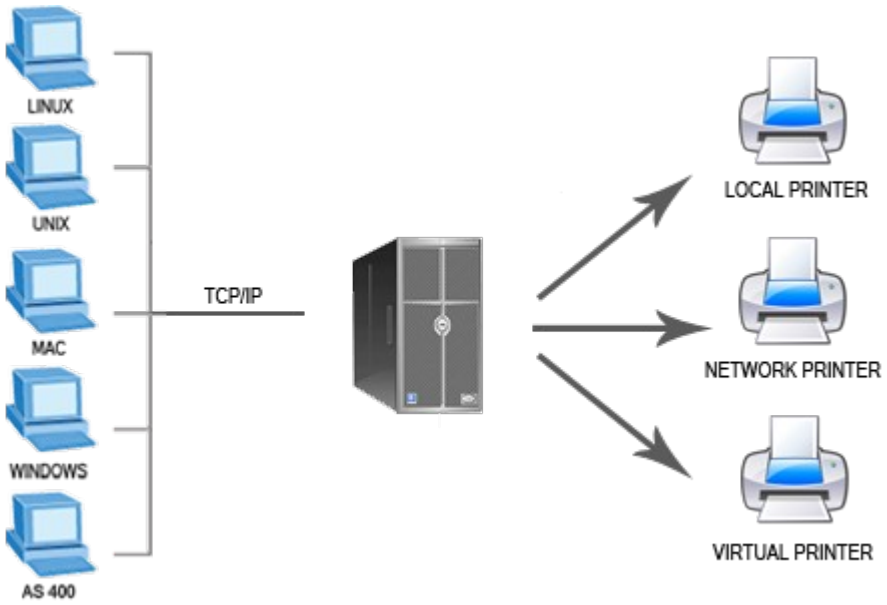
Ejercicio 4: Gestión de impresoras e impresión - Gestión de la hora del sistema



Índice

PARTE I - Impresión.....3
PARTE II - hora del sistema.....3

PARTE I - Impresión

1. Instale el servicio de impresión CUPS y LPD legacy.

```
vagrant@debian:~$ sudo apt-get install cups lpr
sudo: unable to resolve host debian: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  acl avahi-daemon bc colord colord-data cups-browsed cups-client cups-common cups-core-drivers cups-daemon cups-filters cups-filters-core-drivers cups-ipp-utils cups-ppdc cups-server-common fonts-droid-fallback fonts-noto-mono ghostscript gsfonts libavahi-core7
  libavahi-glib1 libcolorhug2 libcupsfilters1 libcupsimage2 libexif12 libfontembed1 libgd3 libgphoto2-6 libgphoto2-l10n libgphoto2-port12 libgs9 libgs9-common libgusb2 libgutenprint-common libgutenprint9 libieee1284-3 libijs-0.35 libjbig2dec0 liblouis-data
  liblouis17 liblouisutdml-bin liblouisutdml-data liblouisutdml8 libnspr4 libnss-mdns libnss3 liboopenjp2-7 libpaper-utils libpaper1 libpolkit-agent-1-0 libpolkit-backend-1-0 libpolkit-gobject-1-0 libpoppler82 libqpdf21 libsane libsane-common libsnmp-base libsnmp30
  libusb-1.0-0 libyaml-0-2 policykit-1 poppler-data poppler-utils printer-driver-gutenprint sane-utils ssl-cert update-inetd
Suggested packages:
  colord-sensor-argyll cups-bsd foomatic-db-compressed-ppds | foomatic-db printer-driver-hpcups hplip cups-pdf smbclient antiword docx2txt imagemagick fonts-noto ghostscript-x libgd-tools gphoto2 gutenprint-locales ooo2dbk rtf2xml snmp-mibs-downloader magicfilter
  | apsfilter fonts-japanese-mincho | fonts-ipafont-mincho fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming fonts-nanum gutenprint-doc unpaper openssl-blacklist
The following NEW packages will be installed:
  acl avahi-daemon bc colord colord-data cups cups-browsed cups-client cups-common cups-core-drivers cups-daemon cups-filters cups-filters-core-drivers cups-ipp-utils cups-ppdc cups-server-common fonts-droid-fallback fonts-noto-mono ghostscript gsfonts
  libavahi-core7 libavahi-glib1 libcolorhug2 libcupsfilters1 libcupsimage2 libexif12 libfontembed1 libgd3 libgphoto2-6 libgphoto2-l10n libgphoto2-port12 libgs9 libgs9-common libgusb2 libgutenprint-common libgutenprint9 libieee1284-3 libijs-0.35 libjbig2dec0
  liblouis-data liblouis17 liblouisutdml-bin liblouisutdml-data liblouisutdml8 libnspr4 libnss-mdns libnss3 liboopenjp2-7 libpaper-utils libpaper1 libpolkit-agent-1-0 libpolkit-backend-1-0 libpolkit-gobject-1-0 libpoppler82 libqpdf21 libsane libsane-common
  libsnmp-base libsnmp30 libusb-1.0-0 libyaml-0-2 lpr policykit-1 poppler-data poppler-utils printer-driver-gutenprint sane-utils ssl-cert update-inetd
0 upgraded, 69 newly installed, 0 to remove and 0 not upgraded.
Need to get 35.4 MB of archives.
After this operation, 131 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Para poder acceder al servidor y a la parte de administración, debemos cambiar un par de parámetros en el fichero de configuración y reiniciar el servicio.

- Permitiríamos conexiones remotas que no estén restringidas a localhost.
- Y permitiríamos el acceso de todo el mundo tanto al acceso como a la parte de administración del servidor.

```
# Only listen for connections from the local machine.
Listen 631
Listen /run/cups/cups.sock

# Show shared printers on the local network.
Browsing On
BrowseLocalProtocols dnssd

# Default authentication type, when authentication is required...
DefaultAuthType Basic

# Web interface setting...
WebInterface Yes

# Restrict access to the server...
<Location />
  Order allow,deny
  Allow from all
</Location>

# Restrict access to the admin pages...
<Location /admin>
  Order allow,deny
  Allow from all
</Location>
```

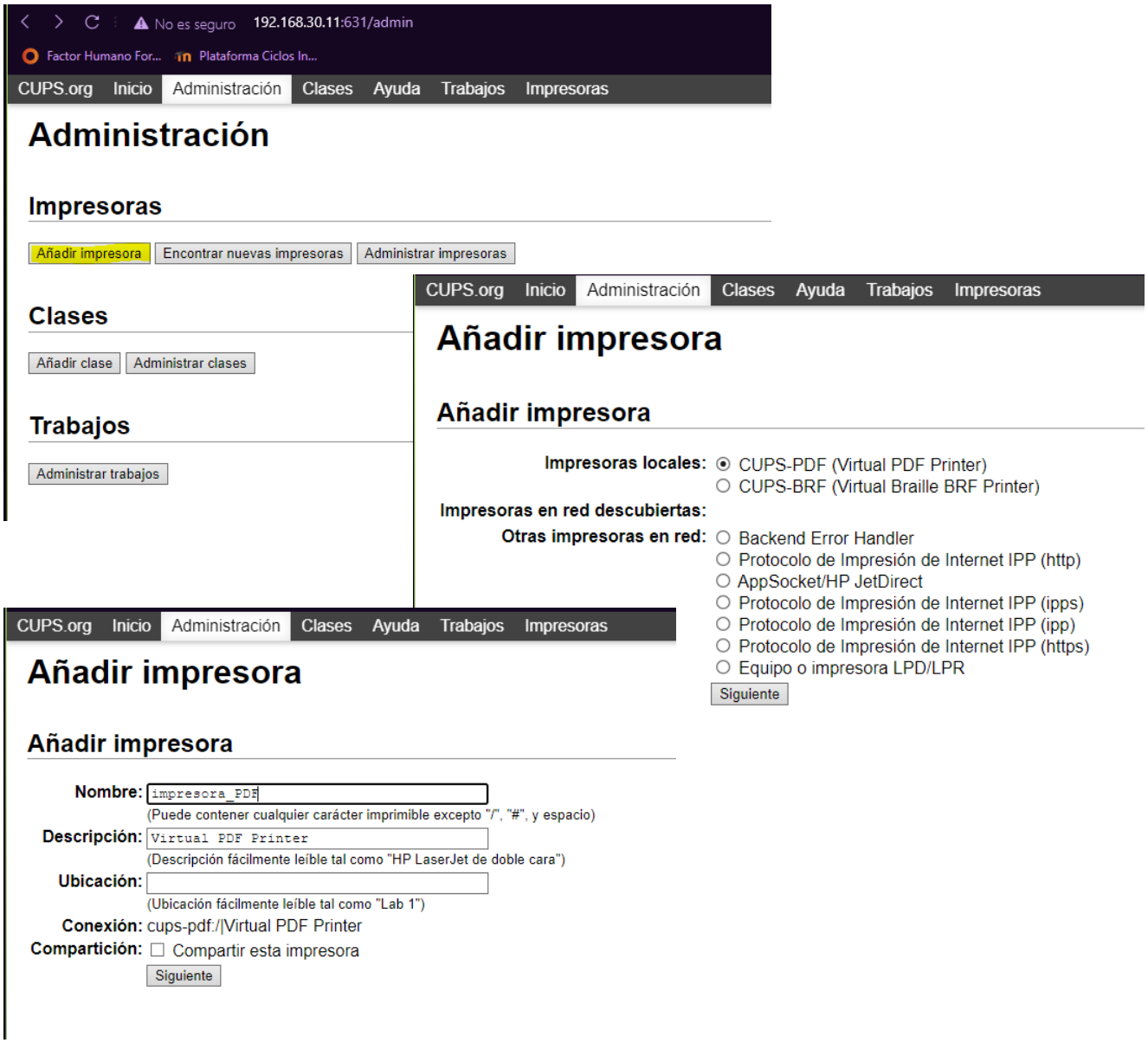
Damos permiso a un usuario para que pueda acceder (lo añadimos al grupo ladmin).

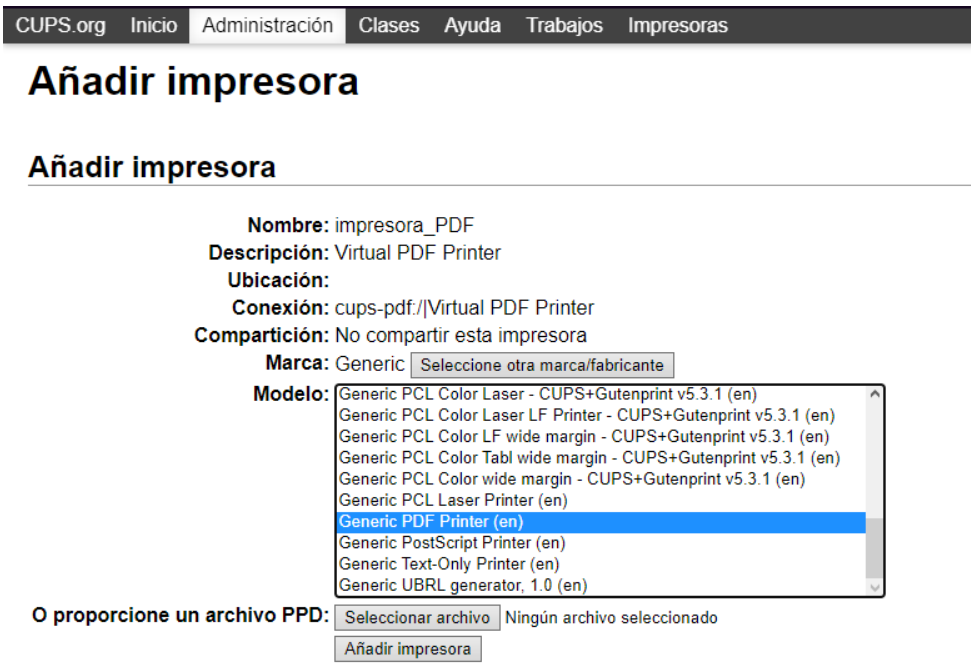
```
vagrant@debian:~$ sudo gpasswd -a vagrant lpadmin
sudo: unable to resolve host debian: Name or service not known
Adding user vagrant to group lpadmin
vagrant@debian:~$ sudo groups vagrant
sudo: unable to resolve host debian: Name or service not known
vagrant : vagrant cdrom floppy sudo audio dip video plugdev netdev bluetooth lpadmin
vagrant@debian:~$
```

2. Instale la impresora CUPS-PDF.

```
vagrant@debian:~$ sudo apt-get install cups-pdf
sudo: unable to resolve host debian: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'printer-driver-cups-pdf' instead of 'cups-pdf'
printer-driver-cups-pdf is already the newest version (3.0.1-5).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vagrant@debian:~$
```

Entramos en la parte de administración de la impresora en https://IP:631/admin y damos click en añadir una impresora.





3. Inicie el servicio CUPS, verifique que se esté ejecutando.

```
vagrant@debian:~$ sudo systemctl start cups
sudo: unable to resolve host debian: Name or service not known
vagrant@debian:~$ sudo systemctl status cups
sudo: unable to resolve host debian: Name or service not known
● cups.service - CUPS Scheduler
   Loaded: loaded (/lib/systemd/system/cups.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2022-03-10 21:02:37 UTC; 6min ago
     Docs: man:cupsd(8)
 Main PID: 10253 (cupsd)
    Tasks: 3 (limit: 1146)
   Memory: 5.7M
   CGroup: /system.slice/cups.service
           ├─10253 /usr/sbin/cupsd -l
           └─10283 /usr/lib/cups/notifier/dbus dbus://

Mar 10 21:02:37 debian systemd[1]: Started CUPS Scheduler.
vagrant@debian:~$
```

4. Enumere todas las impresoras junto con el conjunto de impresoras predeterminado.

```
vagrant@debian:~$ lpstat -s
system default destination: PDF
device for impresora_PDF: cups-pdf:/
device for PDF: cups-pdf:/
vagrant@debian:~$
```

5. Configure la impresora CUPS-PDF como predeterminada.

En este caso, como es la única impresora que se ha configurado, parece que se ha habilitado ella misma por defecto. De todas formas, para seleccionar una por defecto se haría de la siguiente manera.

```
vagrant@debian:~$ lpoptions -d impresora_PDF
copies=1 device-uri=cups-pdf:/ finishings=3 job-cancel-after=10800 job-hold-until=no-hold job-priority=50 job-sheets=none,none marker-change-time=0 number-up=1 printer-commands=none printer-info='Virtual PDF Printer' printer-is-accepting-jobs=true printer-is-shared=fa
lse printer-is-temporary=false printer-location printer-make-and-model='Generic PDF Printer' printer-state=3 printer-state-change-time=1646946453 printer-state-reasons=none printer-type=2101276 printer-uri-supported=ipp://localhost/printers/impresora_PDF
vagrant@debian:~$ lpstat -s
system default destination: impresora_PDF
device for impresora_PDF: cups-pdf:/
device for PDF: cups-pdf:/
```

6. Verifique que se muestre como predeterminado y muestre las opciones de la impresora para esa impresora

```
vagrant@debian:~$ lpoptions
^[[Acopies=1 device-uri=cups-pdf:/ finishings=3 job-cancel-after=10800 job-hold-until=no-hold job-priority=50 job-sheets=none,none marker-change-time=0 number-up=1 printer-commands=none printer-info='Virtual PDF Printer' printer-is-accepting-jobs=true printer-is-share
d=false printer-is-temporary=false printer-location printer-make-and-model='Generic PDF Printer' printer-state=3 printer-state-change-time=1646946453 printer-state-reasons=none printer-type=2101276 printer-uri-supported=ipp://localhost/printers/impresora_PDF
vagrant@debian:~$ lpoptions -l
PageSize/Page Size: *Letter A4 A5 A6 ISOB5 EnvC5 Env10 EnvDL 5x13 EnvC6 Executive B5 B6 Legal EnvMonarch 60x95mm Statement
manualfeed/Manual Feed of Paper: *off on
manualduplex/Manual duplex: *off on
Duplex/Double-Sided Printing: *None DuplexNoTumble DuplexTumble
Resolution/Resolution: 300dpi *600dpi 1200dpi
ret/Resolution Enhancement: dark light medium *notset off
borderless/Borderless Printing: off *on
edgetoedge/Edge to edge: no *yes
joboffset/Job Offset: off *on
lowsupplies/On low supplies: *continue stop
overridea4withletter/Override A4 with Letter: no *yes
vagrant@debian:~$
```

PARTE II - hora del sistema

1. Muestre la fecha y hora actuales en el sistema. Muestra la hora equivalente a UTC. Adelante en una hora la hora del sistema.

```
vagrant@debian:~$ date
Fri 11 Mar 2022 12:25:45 AM UTC
vagrant@debian:~$ sudo date --set "2022-03-11 2:25"
sudo: unable to resolve host debian: Name or service not known
Fri 11 Mar 2022 02:25:00 AM UTC
vagrant@debian:~$
```

2. Muestre la hora del reloj de hardware del sistema. Restablece la hora del sistema en función de a hora del reloj de hardware.

```
vagrant@debian:~$ sudo hwclock
sudo: unable to resolve host debian: Name or service not known
2022-03-11 00:30:52.110603+00:00
vagrant@debian:~$
vagrant@debian:~$ sudo hwclock --systohc
sudo: unable to resolve host debian: Name or service not known
vagrant@debian:~$ date
Fri 11 Mar 2022 12:32:47 AM UTC
vagrant@debian:~$ sudo hwclock
sudo: unable to resolve host debian: Name or service not known
2022-03-11 00:32:50.262386+00:00
vagrant@debian:~$
```

3. Enumera el directorio que es responsable de toda la información de las zonas horarias.

```
vagrant@debian:~$ ls /usr/share/zoneinfo/
Africa      Arctic    Australia CET     Cuba  Eire    Etc     GB      GMT+0 Greenwich Iceland  iso3166.tab Japan            Libya     Mexico  Navajo  Pacific  posix     PST8PDT ROK     Turkey    US  W-SU          Zulu
America     Asia      Brazil    Chile   EET   EST     Europe  GB-Eire GMT-0 Hongkong  Indian   Israel      Kwajalein        localtime MST     NZ      Poland   posixrules right   Singapore UCT     UTC zone1970.tab
Antarctica  Atlantic  Canada    CST6CDT Egypt EST5EDT Factory GMT     GMT0  HST       Iran     Jamaica     leap-seconds.list MET       MST7MDT NZ-CHAT Portugal PRC       ROC     SystemV   Universal WET zone.tab
vagrant@debian:~$
```

4. Instale el servicio ntp. Busque servidores públicos NTP para su región del mundo / país y agréguelos a su configuración. Reinicie el servicio NTP.

```
vagrant@debian:~$ sudo apt-get install ntp
sudo: unable to resolve host debian: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libevent-core-2.1-6 libevent-pthreads-2.1-6 libopts25 sntp
Suggested packages:
  ntp-doc
The following NEW packages will be installed:
  libevent-core-2.1-6 libevent-pthreads-2.1-6 libopts25 ntp sntp
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,174 kB of archives.
After this operation, 2,898 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
  GNU nano 2.3.1

# For more information about this file, see the man pages
# ntp.conf(5), ntp_acc(5), ntp_auth(5), ntp_clock(5), ntp_misc(5), ntp_mon(5).

driftfile /var/lib/ntp/drift

# Permit time synchronization with our time source, but do not
# permit the source to query or modify the service on this system.
restrict default nomodify notrap nopeer noquery

# Permit all access over the loopback interface.  This could
# be tightened as well, but to do so would effect some of
# the administrative functions.
restrict 127.0.0.1
restrict ::1

# Hosts on local network are less restricted.
#restrict 192.168.1.0 mask 255.255.255.0 nomodify notrap

# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (http://www.pool.ntp.org/join.html).
#server 0.centos.pool.ntp.org iburst
#server 1.centos.pool.ntp.org iburst
#server 2.centos.pool.ntp.org iburst
#server 3.centos.pool.ntp.org iburst
server 1.es.pool.ntp.org
server 3.europe.pool.ntp.org
server 0.europe.pool.ntp.org


#broadcast 192.168.1.255 autokey        # broadcast server
#broadcastclient                        # broadcast client
#broadcast 224.0.1.1 autokey            # multicast server
```

5. compruebe el servidor NTP está utilizando dichos servidores para la sincronización, y que la hora está sincronizada correctamente

```
[root@centos7 ~]# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 dns01.masbytes. 147.156.7.26     3 u   36   64    1   43.818   -2.578   0.000
 siberut.ordimat 85.199.214.101   2 u   35   64    1   37.466    1.279   0.000
 185.13.148.71   79.133.44.146    2 u   34   64    1   42.394   -0.045   0.000
[root@centos7 ~]# date
Fri Mar 11 01:35:40 UTC 2022
[root@centos7 ~]#
```